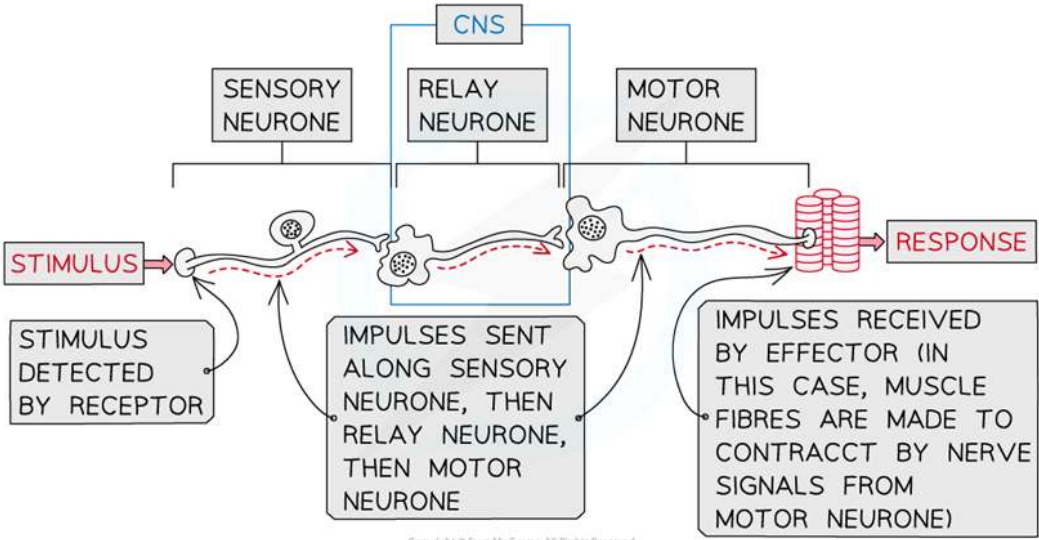
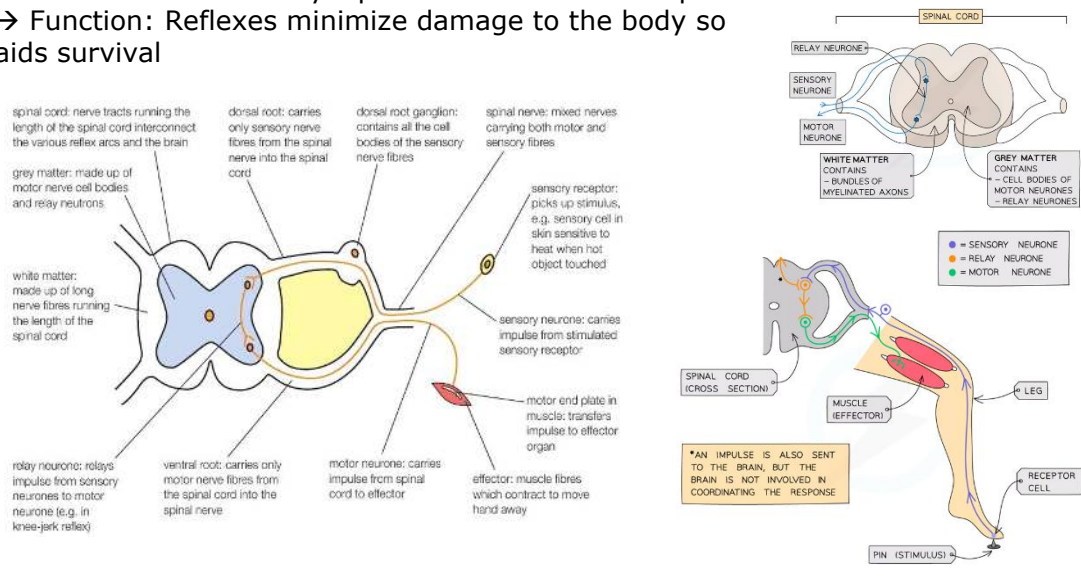


Topic 8 – Coordination, Response and Gene Technology

Candidates will be assessed on their ability to:

8.1	know the structure and function of sensory, relay and motor neurones, including Schwann cells and myelination
	Dendron: part of neurone before cell body Axon: part of neurone after cell body Sensory Neurone: → Structure: - Cell body in the middle → Function: - Carry nerve impulse from receptor organ to CNS Relay Neurone: → Structure: - Bipolar (One dendron and One axon) or Multipolar (With many dendrites) - Located in the grey matter of CNS - Cell body in the middle → Function: - Send impulses from the sensory neurone to motor neurone in reflex arc - Also send the impulse up the spinal cord to the brain (To notify) Motor Neurone: → Structure: - Cell body at the start → Function: - Carry nerve impulse from CNS to effector muscle Myelin Sheath: fatty insulating layer around some vertebrate neurones produced by Schwann cell Node of Ranvier: this is where action potential occurs for myelinated neurones (8.4) (2) Importance of Myelin Sheath 1. Protects nerves from damage 2. Speeds up transmission of nerve impulses fig B The myelin sheath forms a protective, insulating layer around vertebrate nerve fibres. (TS = transverse section, LS = longitudinal section.)

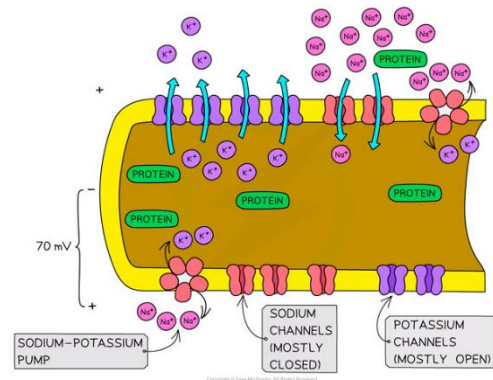
8.2	understand how the nervous system of organisms can cause effectors to respond to a stimulus
	Stimulus → Receptor Organ (Sensory Receptors) → Sensory Neurone → CNS (Brain and Spinal cord) → Motor Neurone → Effector Muscle *Relay neurone is still involved for voluntary responses but now it is from the brain. This is not necessarily mentioned.  Copyright © Save My Exams. All Rights Reserved
8.3	know the structure and function of a spinal reflex arc, including grey matter and white matter of the spinal cord
	The process happens in spinal cord of CNS; however, the brain gets impulse after the action. → Structure: - Receptor cells synapse with sensory neurone (dorsal root) - Sensory neurone synapse with relay neurone in grey matter of spinal arc - Relay neurone synapse with motor neurone (ventral root) - Motor neurone synapse with effector for response → Function: Reflexes minimize damage to the body so aids survival  Labels in diagram include: SPINAL CORD, RELAY NEURONE, SENSORY NEURONE, MOTOR NEURONE, WHITE MATTER (BUNDLES OF MYELINATED AXONS), GREY MATTER (CELL BODIES OF MOTOR NEURONES - RELAY NEURONES), SPINAL CORD (CROSS SECTION), MUSCLE (EFFECTOR), LEG, RECEPTOR CELL, PIN (STIMULUS), and various anatomical structures like dorsal root, dorsal root ganglion, ventral root, and effector. Legend: ● = SENSORY NEURONE ● = RELAY NEURONE ● = MOTOR NEURONE Text box: *AN IMPULSE IS ALSO SENT TO THE BRAIN, BUT THE BRAIN IS NOT INVOLVED IN COORDINATING THE RESPONSE

8.4

understand how a nerve impulse (action potential) is conducted along an axon, including changes in membrane permeability to sodium and potassium ions

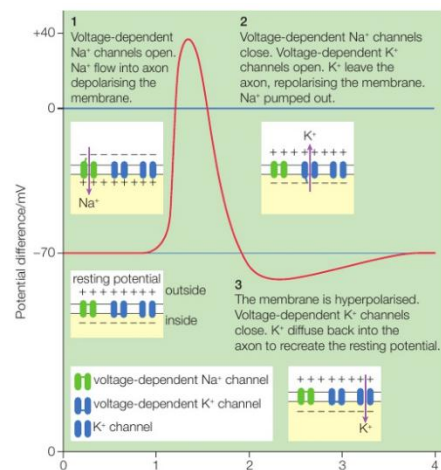
Resting Potential (-70mV):

1. Sodium-potassium pump actively pumps out sodium ions and let potassium ions in (This uses **ATP**)
2. Both (ligand-gated) and (voltage-gated) sodium channels are closed
3. Membrane is relatively permeable to potassium ions
4. Passive diffusion of potassium ions out of the axon
5. So inside of the membrane is negatively charged compared with outside
6. This maintains potential difference



Generator Potential (-55mV):

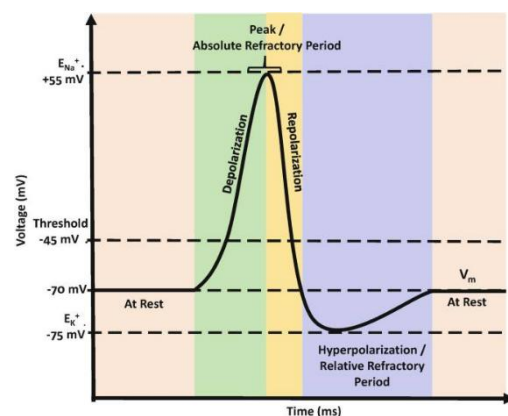
1. Stimulus initiates change in membrane potential
 2. (Ligand-gated) sodium ion channels open
 3. Sodium ions rush in causing membrane potential to become less negative
 4. If above threshold potential, depolarization is initiated
- **Action potential follows all-or-nothing rule** while generator potential does not.
 - No matter how large generator potential is above the threshold potential, action potential will always be (+40mV)
 - **However, bigger stimulus would cause more frequent impulses**



▲ fig B The ionic changes during excitation of an axon result in an action potential.

Action Potential (+40mV):

1. If the generator potential is above threshold potential, voltage-gated sodium channels open
2. There is an influx of sodium ions
3. This **depolarizes** (reverse the membrane potential) the membrane
4. Then voltage-gated sodium channels close
5. Voltage-gated potassium channels open
6. Potassium ions leave the axon, **repolarizing** the membrane
7. As more potassium ions leave the axon, the membrane becomes **hyperpolarized**
8. Voltage-gated potassium channels close
9. Potassium ions diffuse back into the axon to recreate the resting potential



Refractory Period: time it takes for an area of axon membrane to recover after an action potential

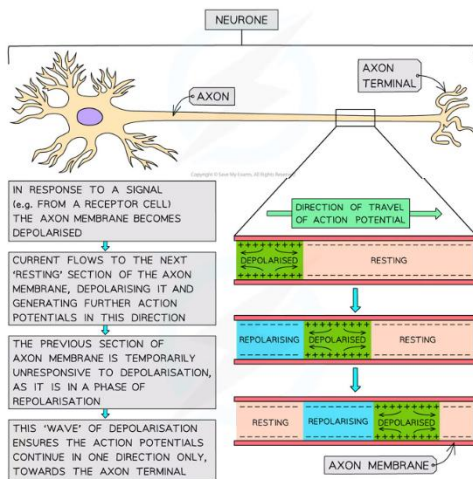
Absolute Refractory Period: time after action potential where sodium channels are completely blocked and resting potential has not been restored

Relative Refractory Period: time when axon will only respond to a much stronger stimulus than before because threshold has been raised (due to hyperpolarization)

(2) Importance of Refractory Period:

1. Limit rate at which impulses may flow along a fibre
2. Ensures impulses flow in **only one direction** along the nerves

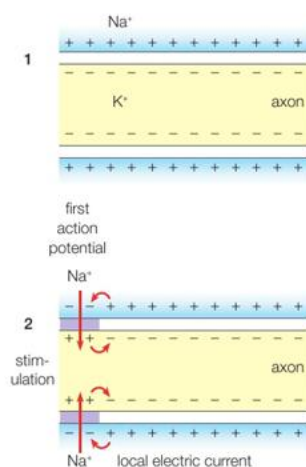
How nerve impulse is transmitted along the axon:



10. Ion movements depolarize membrane in front of action potential sufficiently to cause sodium channels to open

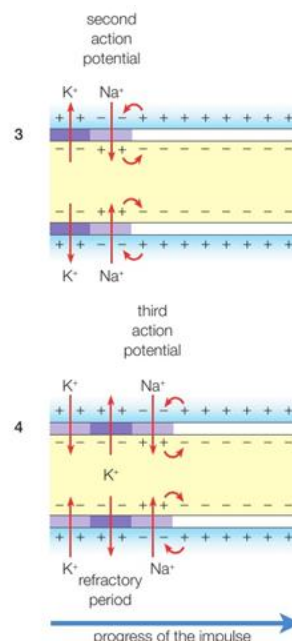
11. This is an example of **positive feedback**.

→ A small depolarization caused by the local currents ahead of the action potential causes the sodium ion channels to open which increases the depolarization.



At resting potential there is positive charge on the outside of the membrane and negative charge on the inside, with high sodium ion concentration outside and high potassium ion concentration inside.

When stimulated, voltage-dependent sodium ion channels open, and sodium ions flow into the axon, depolarising the membrane. Localised electric currents are generated in the membrane.



The potential difference in the membrane adjacent to the first action potential changes. A second action potential is initiated. At the site of the first action potential the voltage-dependent sodium ion channels close and voltage-dependent potassium ion channels open. Potassium ions leave the axon, repolarising the membrane. The membrane becomes hyperpolarised.

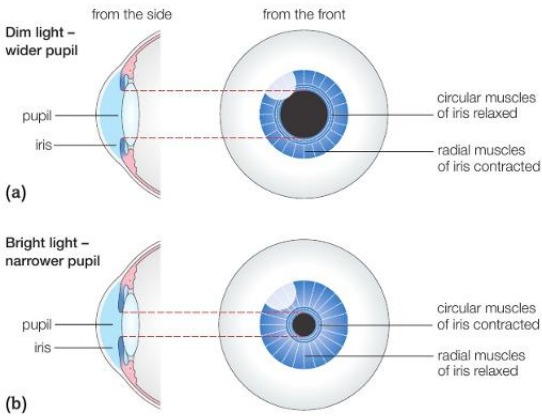
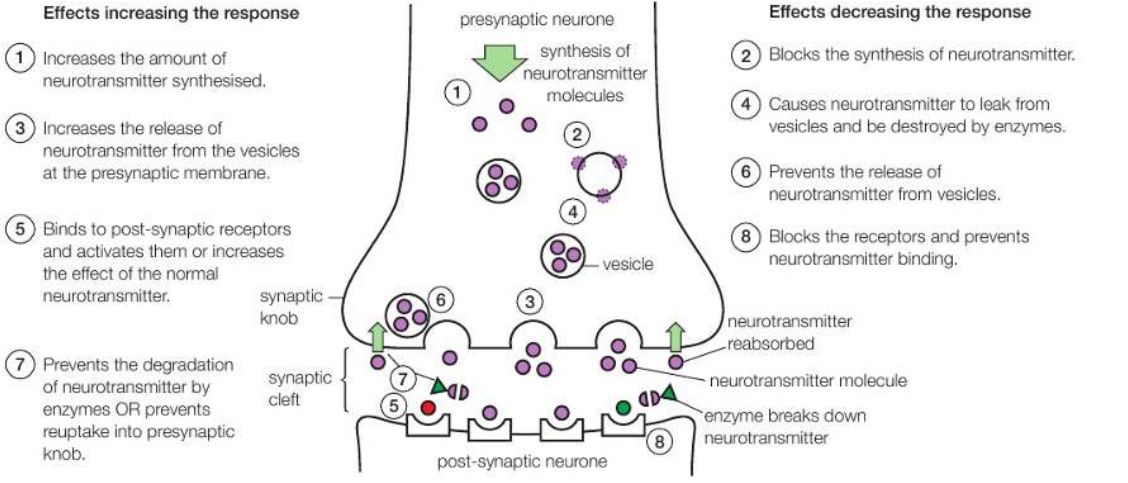
A third action potential is initiated by the second. In this way local electric currents cause the nerve impulse to move along the axon. At the site of the first action potential, potassium ions diffuse back into the axon, restoring the resting potential.

8.5 understand the role of myelination in saltatory conduction

How nerve impulse is transmitted in myelinated neurone:

- Myelin sheath is an insulator
- Nodes of Ranvier have higher number of voltage-gated sodium channels
- Action potentials and depolarization **only** occur at nodes of Ranvier in myelinated neurons
- Therefore, impulses jump from one node of Ranvier to the next and this is called as **Saltatory Conduction**

	<table border="0"> <tr> <td style="vertical-align: top; width: 50%;"> 1 NODE AT REFRACTORY PERIOD: ◦ MEMBRANE BECOMING REPOLARISED ◦ Na^+ CHANNEL PROTEINS CLOSED ◦ K^+ CHANNEL PROTEINS OPEN </td><td style="vertical-align: top; width: 50%;"> 3 NODE BECOMING DEPOLARISED: ◦ MEMBRANE POTENTIAL MOVING TOWARDS THRESHOLD LEVEL ◦ Na^+ CHANNELS STARTING TO OPEN BUT MANY STILL CLOSED ◦ K^+ CHANNEL PROTEINS CLOSED </td></tr> <tr> <td style="vertical-align: top;"> 2 NODE AT ACTION POTENTIAL: ◦ MEMBRANE FULLY DEPOLARISED (+30mV) ◦ ALL Na^+ CHANNEL PROTEINS OPEN ◦ K^+ CHANNEL PROTEINS CLOSED </td><td style="vertical-align: top;"> 4 NODE AT RESTING POTENTIAL: ◦ MEMBRANE POTENTIAL AROUND -70mV ◦ Na^+ CHANNEL PROTEINS CLOSED ◦ K^+ CHANNEL PROTEINS CLOSED </td></tr> </table>	1 NODE AT REFRACTORY PERIOD: ◦ MEMBRANE BECOMING REPOLARISED ◦ Na^+ CHANNEL PROTEINS CLOSED ◦ K^+ CHANNEL PROTEINS OPEN	3 NODE BECOMING DEPOLARISED: ◦ MEMBRANE POTENTIAL MOVING TOWARDS THRESHOLD LEVEL ◦ Na^+ CHANNELS STARTING TO OPEN BUT MANY STILL CLOSED ◦ K^+ CHANNEL PROTEINS CLOSED	2 NODE AT ACTION POTENTIAL: ◦ MEMBRANE FULLY DEPOLARISED (+30mV) ◦ ALL Na^+ CHANNEL PROTEINS OPEN ◦ K^+ CHANNEL PROTEINS CLOSED	4 NODE AT RESTING POTENTIAL: ◦ MEMBRANE POTENTIAL AROUND -70mV ◦ Na^+ CHANNEL PROTEINS CLOSED ◦ K^+ CHANNEL PROTEINS CLOSED
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8.6	(i) know the structure and function of synapses in nerve impulse transmission, including the role of neurotransmitters and acetylcholine (ii) understand how the pupil dilates and contracts				
i	Structure: Function: 1. Arrival of impulse at synaptic knob opens voltage-gated calcium ion channels open in presynaptic membrane causing calcium ions to enter 2. This causes vesicles of neurotransmitter to fuse with presynaptic membrane 3. Neurotransmitters are released into synaptic cleft through exocytosis 4. Neurotransmitter diffuses across and binds to receptors on postsynaptic membrane 5. This causes ligand-gated sodium channels to open, resulting in depolarization leading to action potential in postsynaptic neurone				

	(2) Common Neurotransmitters 1. Acetylcholine - Cholinergic nerves use acetylcholine as their transmitter - Acetylcholinesterase enzyme is used to hydrolyze acetylcholine into acetate and choline once they bind to specific receptors and triggered a post-synaptic potential → This prevents ACh binding again to the receptor and also help to recycle by absorbing back into the pre-synaptic membrane - Parasympathetic nerves use acetylcholine as their transmitter 2. Noradrenaline - Adrenergic nerves use noradrenaline as their transmitter - Sympathetic nerves use noradrenaline as their transmitter
ii	Cranial Reflex: Controlling amount of light that enters the eye Light -> Sensory cells in Retina -> Optic nerves -> Brain (midbrain where control centres are located) -> Oculomotor nerves -> Effectors (Iris muscles) Circular Muscles are arranged in concentric rings around pupil and Radial Muscles run radially and they work antagonistically Bright Light: - Circular muscles contract - Radial muscles relax Dim Light: - Circular muscles relax - Radial muscles contracts *BCC (Bright Circular Contracts) 
8.7	understand how the effects of drugs can be caused by their influence on nerve impulse transmission, illustrated by nicotine, lidocaine and cobra venom alpha toxin, the use of L-DOPA in the treatment of Parkinson's disease and the action of MDMA (ecstasy)
	 Effects increasing the response - Increases the amount of neurotransmitter synthesised. - Increases the release of neurotransmitter from the vesicles at the presynaptic membrane. - Binds to post-synaptic receptors and activates them or increases the effect of the normal neurotransmitter. - Prevents the degradation of neurotransmitter by enzymes OR prevents reuptake into presynaptic knob. presynaptic neurone synthesis of neurotransmitter molecules vesicle synaptic knob synaptic cleft post-synaptic neurone neurotransmitter reabsorbed neurotransmitter molecule enzyme breaks down neurotransmitter Effects decreasing the response - Blocks the synthesis of neurotransmitter. - Causes neurotransmitter to leak from vesicles and be destroyed by enzymes. - Prevents the release of neurotransmitter from vesicles. - Blocks the receptors and prevents neurotransmitter binding.

	Nicotine: - Found in tobacco - Mimics effect of acetylcholine and bind to nicotinic receptors - Triggers action potential but receptor remains unresponsive to more stimulation for some time - Heart rate and blood pressure increases - Also triggers release of dopamine which associates with pleasure sensation - At low dose, nicotine is a stimulant - At high dose, it blocks acetylcholine receptors and can kill Lidocaine: - Used in local anaesthetic - Block voltage-gated sodium channels to prevent action potential in sensory nerves so no pain sensation - Also used to prevent heart arrhythmias → Block sodium channels which raise depolarization threshold and this reduces early or extra action potentials from the pacemaker region that can cause arrhythmias Cobra venom (alpha-cobra toxin) - Bind irreversibly to acetylcholine receptors in postsynaptic membrane and neuromuscular junctions → Muscles are not stimulated to contract and gradually the person affected become paralyzed → If it reaches to respiratory muscles, death will occur - At very low dose, it is used as muscle relaxant in severe asthma attacks L-DOPA - Used for Parkinson's disease treatment - In Parkinson's disease, dopamine producing cells in the motor cortex die - L-DOPA crosses the blood-brain barrier while dopamine cannot - In the brain, L-DOPA is converted to dopamine - This restores control of muscle movement MDMA (Ecstasy) - MDMA increases level of serotonin in the brain - This is done by (2) ways 1. Serotonin reuptake from presynaptic knob is inhibited 2. MDMA crosses blood-brain barrier and binds to receptors in neurone to stimulate the release of serotonin in the brain - High serotonin level also stimulates release of more dopamine - Increase heart rate and blood pressure - No desire to drink - Affect hypothalamus to secrete more antidiuretic hormone (ADH) so conserve water → Higher ADH so more water reabsorbed which leads to dilution of blood → Diluted blood has a higher water potential than brain cells so water moves into brain cells by osmosis; therefore, brain swelling occurs
8.8	understand how the nervous systems of organisms can detect stimuli with reference to rods in the retina of mammals, the roles of rhodopsin, opsin, retinal, sodium ions, cation channels and hyperpolarization of rod cells in forming action potentials in the optic neurones

	Cells: → Structure: - Contains photoreceptor called Rhodopsin → Rhodopsin is made of opsin and retinal → Retinal has cis-trans isomer → Dark: cis form → Light: trans form → Function: Dark: - Rods are not stimulated; Exists as rhodopsin - Sodium-potassium pump moves sodium ions out - Membrane of rod cells are permeable to sodium ions due to sodium channels → Sodium ions diffuse back into the cell through open sodium channels Light: - Rhodopsin is photoreceptor and absorbs light energy - Cis retinal and opsin is converted into trans retinal and opsin - Sodium channels in the membrane are closed - Sodium-potassium pump is still actively moving sodium ions out → Membrane becomes hyperpolarized as the inside the membrane becomes very negative (which raises threshold potential) → This is the generator potential in the rod cell. (Size of the generator potential depends on amount of bleached rhodopsin which means amount of light) → If generator potential $\geq$ threshold potential, neurotransmitters released to bipolar cells → This causes action potential in bipolar cell to action potential in sensory neurone
8.9	understand what is meant by the term <i>habituation</i>
	Habituation: Ignoring an innate response to a frequently repeated stimulus → This filters out what is important and what is not important to react Process: 1. Calcium ion channels in presynaptic membrane become less responsive with repeated stimulation 2. Fewer calcium channels opens so fewer calcium ions cross into presynaptic knob 3. Fewer vesicles fuse with the presynaptic membrane to discharge their neurotransmitters 4. Less neurotransmitter available to bind to postsynaptic membrane so postsynaptic excitatory potential is not high enough to trigger an action potential so no response fig 8 Repeated stimulation of the siphon of <i>Aplysia</i> results in habituation (a) and (b). This is a simple form of learning whereby the sea slug stops responding to the stimulus as a result of changes in the response of the synapse, as shown in (c).

RECOMMENDED ADDITIONAL PRACTICAL

Investigate habituation to a stimulus.

- 8.10** know that the mammalian nervous system consists of the central and peripheral nervous systems

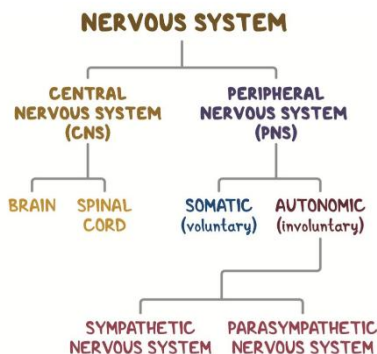


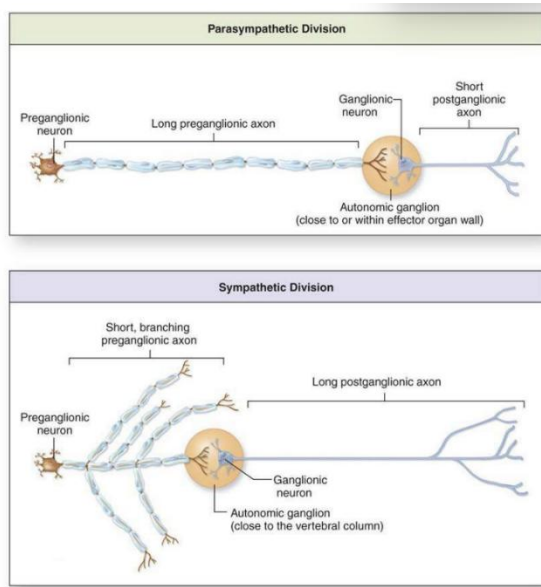
Figure 50.1 Organization of the nervous system.

Parasympathetic Nerves:

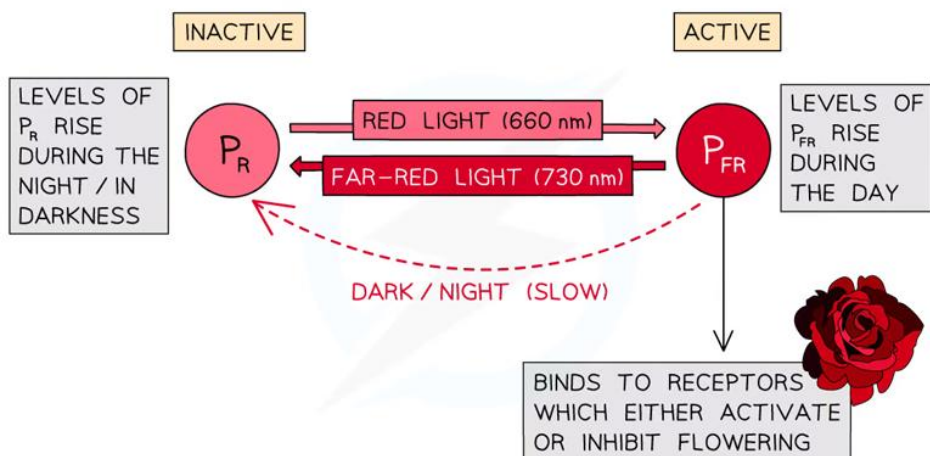
- Preganglionic fibres → Longer
- Postganglionic fibres → Shorter
- So, ganglia are closer to effector
- Produces **acetylcholine** at synapses
- Slower, calming or inhibitory effect on organ

Sympathetic Nerves:

- Preganglionic fibres → Shorter
- Postganglionic fibres → Longer
- So, ganglia are closer to CNS
- Produces **noradrenaline** at synapses
- Rapid response in target organ



- 8.11** understand how phytochrome, auxin (IAA) and gibberellins bring about responses in plants, including their effects on transcription

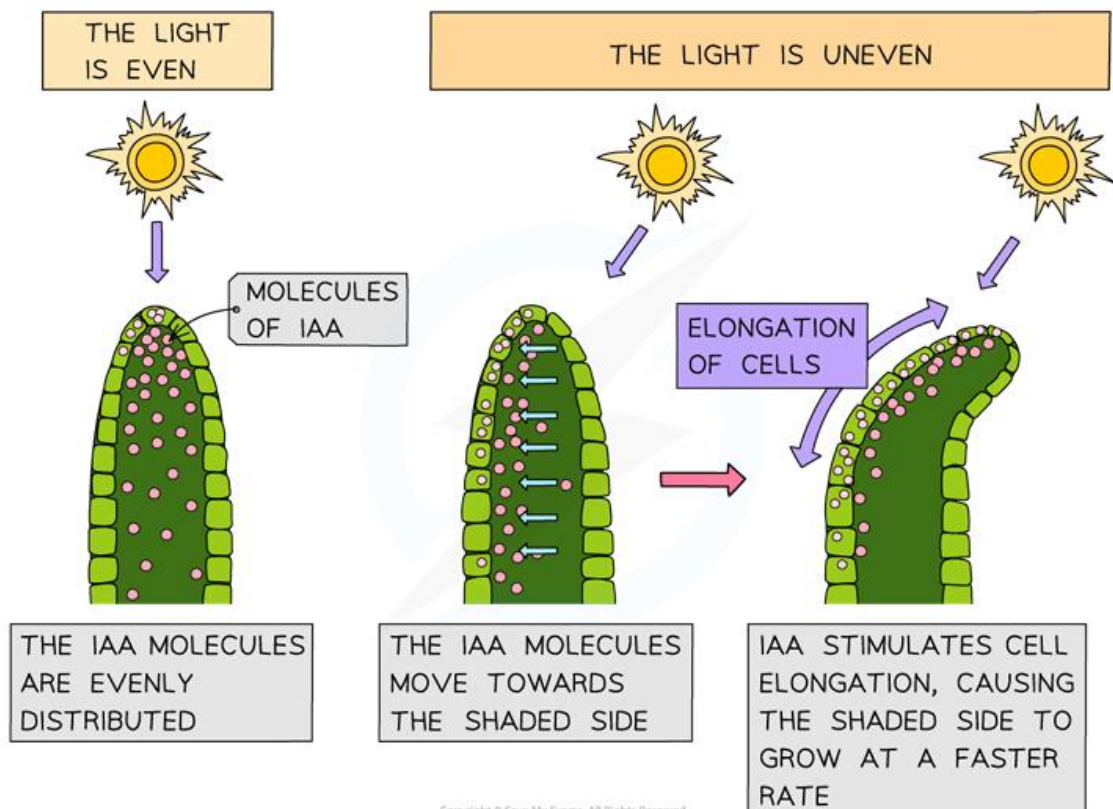


Phytochrome

- There are **(2) forms** of phytochrome: **P_r** (Phytochrome Red) and **P_{fr}** (Phytochrome Far Red)
- → These two pigments differ in their absorption peak.

Mechanism of Phytochromes:

- P_{fr} → absorbs far red light (730nm) → P_r (inactive)
- P_r → absorbs red light (660nm) → P_{fr} (active form)
- Full sunlight contains more red light than far red light
→ In light, all phytochrome is P_{fr}
→ In the dark, P_{fr} spontaneously converts back to P_r
- Active form P_{fr} can activate/repress specific gene expression linked to flowering as P_{fr} is a transcription factor.
- In **short-day** plants, P_{fr} **inhibits** flowering
→ So, for short-day plants to flower, P_{fr} must be low and P_r must be high
- In **long-day** plants, P_{fr} **promotes** flowering
→ So, for long-day plants to flower, P_{fr} must be high and P_r must be low
- This is called **photoperiodism** where relative duration of light and dark periods is important for plant flowering

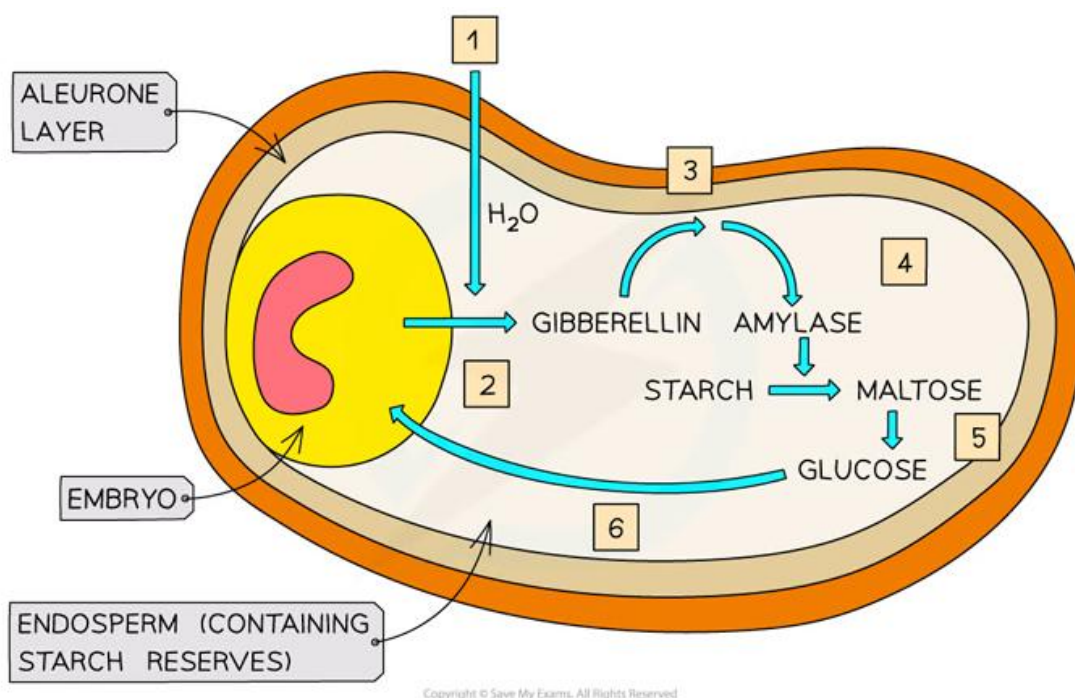


Auxin (Indoleacetic Acid [IAA])

- Phototropism is caused by auxins
- Auxin is transported down dark side and less on bright side
- Auxin stimulates elongation on shaded side to give curvature
- This causes plant to go towards light

Mechanism of Auxin:

1. Auxin affects hydrogen ion pumps
2. Auxin causes intracellular reserves of hydrogen ions to increase which lowers pH on dark side
3. Lower pH activates enzymes
4. These disrupts hydrogen cross-linkage in cell wall, increasing osmotic gradient due to movement of solutes
5. Water enters cell and increased turgor pressure leads to swelling of cell so cell wall stretches allowing cell to elongate and expand



Gibberellin

1. Break dormancy in seeds and in germination

1. Seed absorbs water and swells which activates embryo.
2. Embryo secretes gibberellin that diffuses to aleurone layer
3. Gibberellin stimulates aleurone layer to produce amylase (starch digesting enzyme) that diffuses into endosperm and breaks down food stores to provide embryo with materials for respiration and growth.

Mechanism of Gibberellin

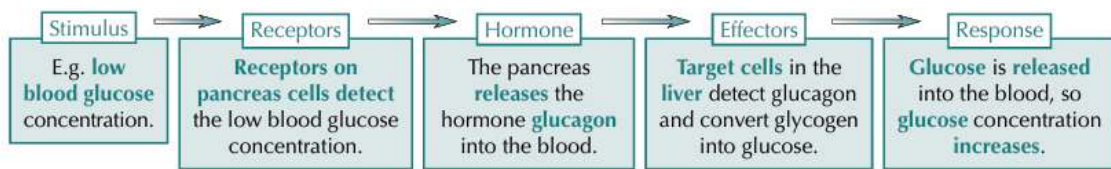
1. Gibberellin enters the nucleus and bind to receptor
2. Gibberellin acts as transcription factor
3. Amylase gene is activated
4. Allowing amylase gene transcription to produce amylase
5. Amylase hydrolyses starch into glucose

8.12 CORE PRACTICAL 18

Investigate the production of amylase in germinating cereal grains.

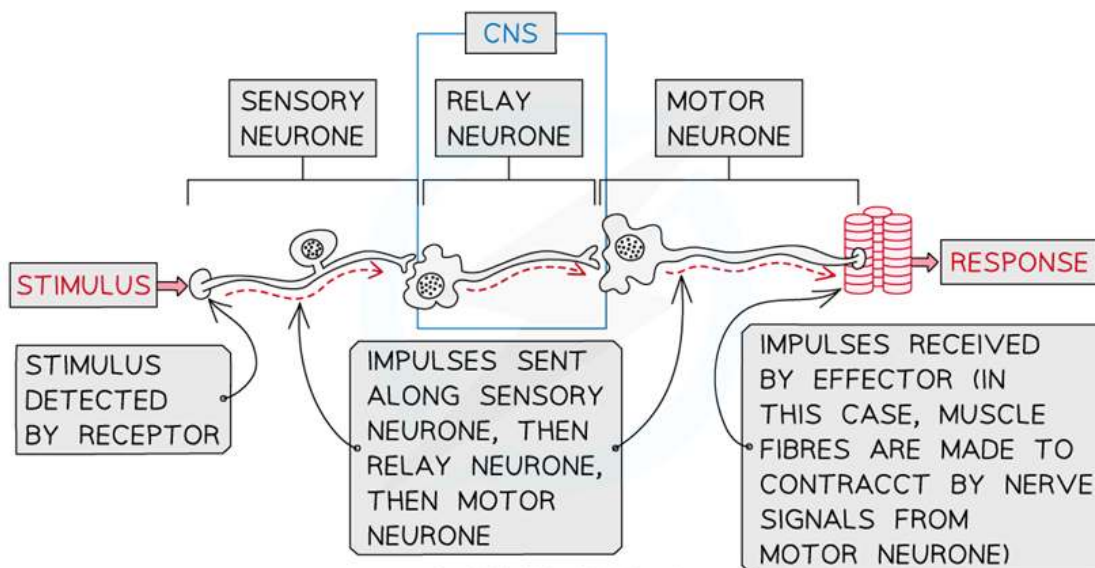
8.13 understand how coordination in animals is brought about through nervous and hormonal control

Hormonal Control System



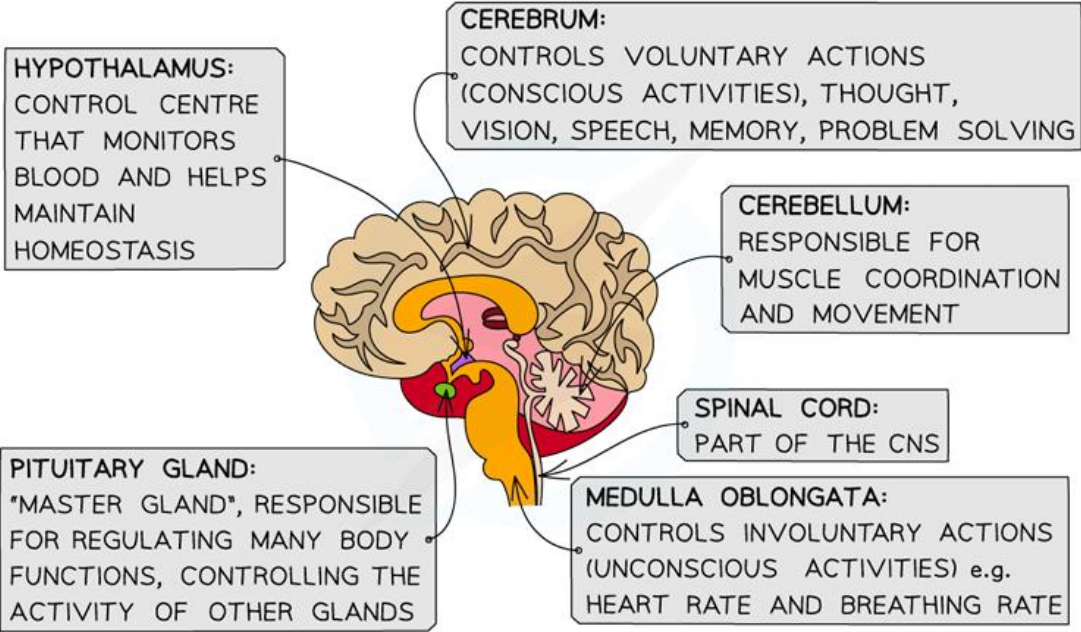
- Slow
- Long lasting effect
- Use chemicals where they are transported by blood plasma

Nervous Control System



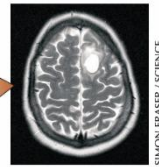
- Rapid response
- Short-lived response
- Use electrical impulses carried by nerve fibres

8.14 know the location and main functions of the cerebral hemispheres, hypothalamus, pituitary gland, cerebellum and medulla oblongata of the human brain

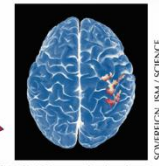
	 HYPOTHALAMUS: CONTROL CENTRE THAT MONITORS BLOOD AND HELPS MAINTAIN HOMEOSTASIS CEREBRUM: CONTROLS VOLUNTARY ACTIONS (CONSCIOUS ACTIVITIES), THOUGHT, VISION, SPEECH, MEMORY, PROBLEM SOLVING CEREBELLUM: RESPONSIBLE FOR MUSCLE COORDINATION AND MOVEMENT PITUITARY GLAND: "MASTER GLAND", RESPONSIBLE FOR REGULATING MANY BODY FUNCTIONS, CONTROLLING THE ACTIVITY OF OTHER GLANDS SPINAL CORD: PART OF THE CNS MEDULLA OBLONGATA: CONTROLS INVOLUNTARY ACTIONS (UNCONSCIOUS ACTIVITIES) e.g. HEART RATE AND BREATHING RATE Copyright © Save My Exams. All Rights Reserved Cerebral Hemisphere (Cerebellum): - Control voluntary behavior - Site of intelligence - Learning, thinking, imaging, motor skills (cerebral cortex) Medulla Oblongata: - Control heart rate and breathing rate Hypothalamus: - Homeostasis: Thermoregulation, Osmoregulation - Controls pituitary gland Cerebrum: - Coordination of voluntary movements, posture and balance
8.15	understand how magnetic resonance imaging (MRI), functional magnetic resonance imaging (fMRI), positron emission tomography (PET) and computed tomography (CT) are used in medical diagnosis and the investigation of brain structure and function



This CT scan looks up at the brain from below. The white area is heavy bleeding due to a blood vessel ruptured after a stroke.



This MRI scan looks down on the brain from above. The white area is a tumour.



This fMRI scan looks down on the brain from above. The red area is active when the person moves their left hand. The right side's active because it controls the left side of your body.

Computed Tomography (CT)

- Involves thousands of tiny beams of X-rays
 - Each beam is reduced in strength by the density of the tissue it passes through
 - X-rays which pass through the tissue are detected
- Can detect major structures but not fine structural details
 - Cannot be used to show how areas of brain are used or change during different activities but can link brain function with behavior
- Sometimes special dyes are injected into the blood or tissue which stop X-rays from passing, so they show up more clearly in the scan

- X-ray is used so radiation risk exists

Magnetic Resonance Imaging (MRI)

- Produce images that show much finer details than CT scans
- Produced using magnetic fields and radio waves to image the soft tissues
- Hydrogen atoms are most common imaged element as body is made of water (H_2O) and hydrogen atoms produce a strong MRI signal
 - Different tissues respond differently to the magnetic field from the radio waves depending on amount of water in the structure

- + No ionizing radiation
- + Good soft-tissue contrast
- + Use magnetic field so safe for pregnant women (not all individuals)
- Enclosed space
- Noisy

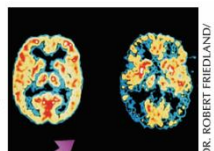
Functional Magnetic Resonance Imaging (fMRI)

- Allow to watch different areas of the brain in action while people perform different tasks
- Monitor the absorption of oxygen in different brain areas
 - Deoxyhaemoglobin absorbs the radio-wave signal and later re-emits it, whereas oxyhaemoglobin does not
- Blood flow to active area of the brain increases as oxygen is needed for aerobic respiration so less signal is absorbed (means more is reflected and can be seen on the scan)

Positron Emission Tomography (PET)

1. Isotope that emits positron is incorporated into glucose
 2. More active neurons with increased respiration will require increased supply of glucose
 3. Positron is emitted from glucose which produce gamma rays that are detected and are converted into an image
- + Show parts of brain are active
 - Exposure to gamma radiation
 - Expensive

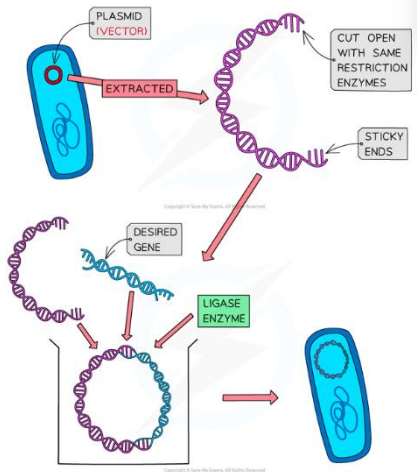
This PET scan shows a normal brain (left) and one from a person with Alzheimer's (right) — you can see the reduction in function in the brain with Alzheimer's.



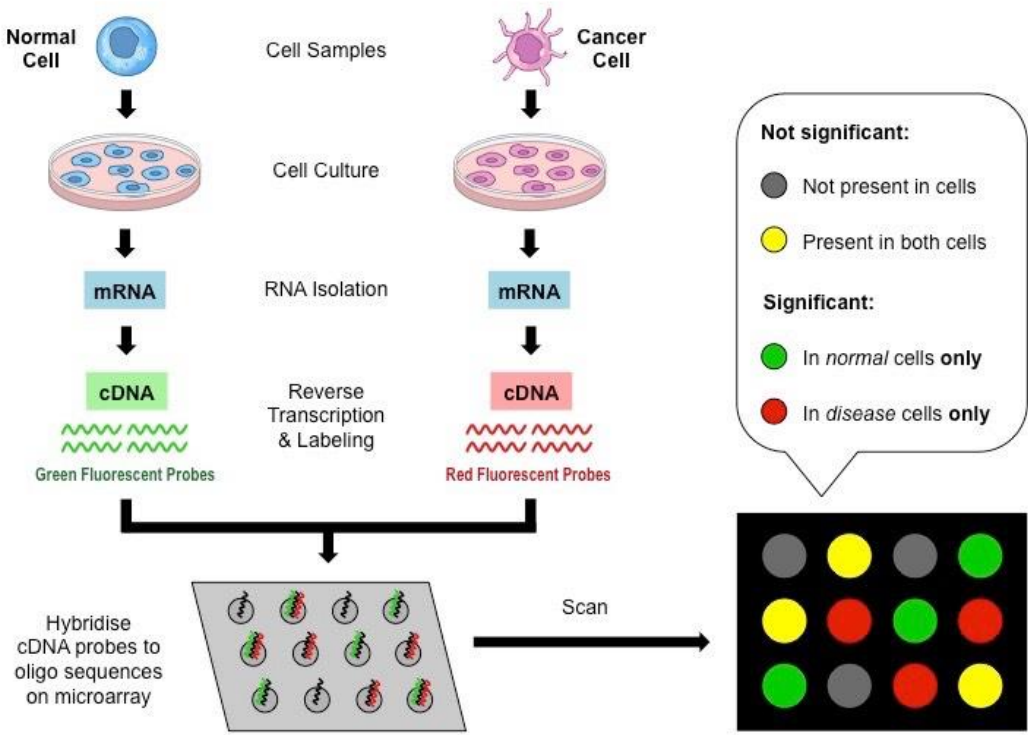
8.16 understand how imbalances in certain naturally-occurring brain chemicals can contribute to ill health, including dopamine in Parkinson's disease and serotonin in depression, and to the development of new drugs

Parkinson's Disease

- Dopamine producing nerve cells in substantia nigra of the midbrain region dies

	- Fewer dopamine molecules binds to ligand gated sodium channels on the postsynaptic membrane - Initiating fewer action potentials in postsynaptic neurone - Fewer impulses are sent to the parts of the brain controlling motor functions - Treatments: - Levodopa (L-DOPA): precursor to dopamine and cross blood-brain barrier to travel into the brain and convert into dopamine - Dopamine agonists: bind to dopamine receptors and mimic effect of dopamine - Monoamine oxidase B (MAOB) inhibitors: inhibit enzyme monoamine oxidase B that breaks down dopamine in brain synapses Depression - Depression is linked to low level of serotonin - Serotonin transmits nerve impulses across synapses in the parts of the brain that control mood - Treatments: - Selective Serotonin Reuptake Inhibitors (SSRIs): inhibit reuptake of serotonin in the presynaptic membrane so more serotonin left in synaptic clefts which allows more impulses to travel along postsynaptic neurone - Tricyclic Antidepressants (TCAs): increase levels of serotonin and noradrenaline in the brain - Monoamine oxidase Inhibitors: inhibit enzymes that cause breakdown of neurotransmitters in the synapse of the brain
8.17	know how drugs can be produced using genetically modified organisms (plants, animals and microorganisms)
	General Genetic Modification Process: 1. Isolate the desired gene 2. Cut the desired DNA sequence and vector's DNA/plasmid using same restriction endonuclease 3. Use ligase enzyme to join sticky ends of the desired gene and vector's DNA/plasmid 4. Method of insertion (8.19) 5. Use bio fermenters for bacteria to grow, greenhouse for plants and physical containment for animals 
8.18	understand how recombinant DNA can be produced, including the roles of restriction endonucleases and DNA ligase
	Restriction Endonuclease: enzymes that cut DNA through sugar-phosphate backbone in an uneven way leaving exposed single stranded sequences of bases known as 'sticky ends' Ligase Enzyme: enzymes that catalyze formation of phosphodiester bonds between the sugar-phosphate backbone of desired gene and vector DNA to produce recombinant plasmid / DNA

8.19	understand how recombinant DNA can be inserted into other cells
	Method of insertion → Vector for Microorganism - the recombinant plasmid taken up by bacteria due to heat shock → Vectors for Plant and Animals - Gene guns - Liposome - Microinjection - Infection through non-lytic virus - Infection through Agrobacterium (Plant) - Infection through Adenovirus (Human)
8.20	know how microarrays can be used to identify active genes
	Microarrays allow identification of active genes and activity of many genes can be analyzed in a sample. This allows to collect information about genetic differences between many individuals Procedure for DNA microarray: 1. Sample Collection: Collect mRNA samples from reference sample (normal cell) and experimental sample (cancer cell) 2. Reverse Transcription: Use reverse transcriptase enzymes to convert the collected mRNA into cDNA (complementary DNA). 3. Fluorescent Labelling: Apply fluorescent labels to the cDNA samples. → Reference sample: Green fluorescent dye → Experimental sample: Red fluorescent dye 4. Hybridisation: Mix the labelled DNA samples and apply them to the microarray slide so the cDNA sequences will bind to their matching DNA probes fixed on the slide's spots. 5. Scanning and Analysis: The slide is scanned to measure the fluorescent light emitted from each spot. Yellow (Red + Green): Both samples are expressing the gene equally. Red: The experimental sample is expressing the gene more than the control. Green: The experimental sample is expressing the gene less than the control.

	
8.21	understand what is meant by the term <i>bioinformatics</i>
	Bioinformatics develops algorithms to identify gene sequences and analyses the data to identify differences such as mutation between healthy and disease having individuals
8.22	understand the risks and benefits associated with the use of genetically modified organisms
	Benefits of Genetically Modified Organisms: 1. Higher yields / Less damage by pests 2. More appealing color / Useful feature 3. Genetically engineered foods have longer shelf life 4. Improved nutritional content 5. Medication provided through genetically engineered foods 6. Can help prevent malnutrition 7. Reduced use of herbicides and pesticides 8. Withstand environmental conditions Risks of Genetically Modified Organisms: 1. Increasing demand in land use 2. Reducing land available to grow other crops 3. Increasing monoculture which reduce biodiversity 4. Unknown consequences of foreign gene expression 5. Cross-breeding with other species 6. Allows production of superpests 7. Horizontal gene transfer to other organisms (through conjugation) How risks can be reduced: - Grow genetically engineered plants in greenhouse

Notice

- In this booklet, mark scheme points from past paper questions (mostly 2020-2023 and a few from 2010-2019) are included for each specification point to connect the syllabus and the past paper questions.
- For less frequently examined specification points, Pearson Edexcel IAL textbook and Save My Exams (available at [savemyexams.com](https://www.savemyexams.com)) notes have been the main reference of this booklet.
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